Data Dictionary – Climate & Environmental Star Schema

This data model follows a **star schema design** with multiple fact tables at different granularities, supported by shared dimensions. The structure is designed to separate **raw daily observations**, **monthly trend analysis**, and **contextual environmental indicators**.

🧱 Dimension Tables

Dim_Calendar

Grain: One row per date (monthly granularity used as the primary key for facts)

Column	Type	Description
Date	Date	Reference date (first day of month or year, depending on fact)
Year	Integer	Calendar year
Month	Integer	Calendar month (1–12)
YearMonth	Text	Time key in YYYY-MM format
MonthName	Text	Month name
Season	Text	Climatic season (Winter, Spring, Summer, Fall)

Purpose:
Shared time dimension for all fact tables.

Dim_Station

Grain: One row per weather station

Column	Type	Description
ClimateID	Text	Official station identifier (alphanumeric)
StationName	Text	Station name
Latitude	Decimal	Geographic latitude

Longitude	Decimal	Geographic longitude
ProvinceCode	Text	Province code (NS, NB, PE, NL)

Purpose:
Spatial filtering and aggregation for station-level facts.

Dim_Region

Grain: One row per province or region

Column	Type	Description
RegionID	Integer	Surrogate key
RegionName	Text	Province or region name
ProvinceCode	Text	Official province or regional code (e.g., NS, ATL)

Purpose:
Regional aggregation and contextual analysis.

📊 Fact Tables

Fact_Weather_Daily

Grain: Station × Day
Source: ECCC – Daily Climate Observations

Column	Type	Description
ClimateID	Text	FK → Dim_Station
DateTime	DateTime	Date of the daily observation
MaxTemp_C	Decimal	Daily maximum temperature (°C)
MinTemp_C	Decimal	Daily minimum temperature (°C)

MeanTemp_C	Decimal	Daily mean temperature (°C)
HeatDegDays	Decimal	Heating Degree Days
CoolDegDays	Decimal	Cooling Degree Days
TotalRain_mm	Decimal	Daily rainfall (mm)
TotalSnow_cm	Decimal	Daily snowfall (cm)
TotalPrecip_mm	Decimal	Total daily precipitation (mm)
SnowOnGround_cm	Decimal	Snow on ground (cm)
MaxGustDir_deg	Decimal	Direction of maximum wind gust (degrees)
MaxGustSpeed_kmh	Decimal	Speed of maximum wind gust (km/h)

Purpose:
Base table used for:

- temperature anomaly calculations
- extreme weather event detection
- derivation of aggregated monthly metrics

⚠️ **Not used directly for long-term BI trend visuals.**

Fact_Temperature_Monthly

Grain: Station × Month
Source: Aggregated from Fact_Weather_Daily (Python)

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Column	Type	Description
ClimateID	Text	FK → Dim_Station
Date	Date	FK → Dim_Calendar (monthly)
MeanMonthlyTemp_C	Decimal	Mean monthly temperature (°C)

Purpose:

- Long-term temperature trend analysis

- Dual-axis visualization with precipitation
- Year-over-year comparisons

Fact_Precipitation_Monthly

Grain: Station × Month
Source: ECCC – Monthly Climate Observations

Column	Type	Description
ClimateID	Text	FK → Dim_Station
Date	Date	FK → Dim_Calendar (monthly)
Precipitation_mm	Decimal	Total monthly precipitation (mm)
ProvinceCode	Text	FK → Dim_Region

Purpose:

- Long-term precipitation trends
- Dual-axis visualization with temperature
- Regional comparisons

Fact_ExtremeEvents

Grain: Station × Period × Event Type
Source: Derived from Fact_Weather_Daily (Python)

Column	Type	Description
ClimateID	Text	FK → Dim_Station
Date	Date	FK → Dim_Calendar
EventType	Text	Extreme event type (e.g., Heavy Precipitation, High Wind)
EventCount	Integer	Number of events in the period

Purpose:
Supports analysis of trends in extreme weather frequency, addressing the project question on measurable changes in extreme events at weather stations.

Fact_AQI

Grain: Region × Month
Source: Air quality datasets (contextual)

Column	Type	Description
RegionCode	Text	FK → Dim_Region
Date	Date	FK → Dim_Calendar
AQI_Value	Decimal	Average Air Quality Index value
AQI_Category	Text	Qualitative category (Low, Moderate, High Risk)

Purpose:
Provides contextual air quality information to support environmental interpretation.

⚠️ Calendar relationship is kept **inactive** to avoid ambiguous filter paths.

Fact_GHG

Grain: Region × Year
Source: Government greenhouse gas emission statistics

Column	Type	Description
RegionCode	Text	FK → Dim_Region
Year	Integer	FK → Dim_Calendar
GHG_Emissions	Decimal	Total greenhouse gas emissions
Unit	Text	Measurement unit (e.g., Mt CO_2e)

Purpose:
Contextual analysis of long-term emissions and industrial activity trends.

🧭 **Model Notes (Schema Walkthrough Context)**

- Fact tables intentionally operate at **different granularities** (daily, monthly, annual).
- **Monthly facts** support trend analysis and BI visualization.
- **Daily weather data** is retained for analytical derivations (anomalies and extreme events).
- Contextual facts (AQI, GHG) are modeled separately to avoid interference with core meteorological trends.
- Differences in station or temporal coverage reflect **real-world data availability** and are expected.